AI-POWERED SMART WASTE SEGREGATION SYSTEM

SOFTWARE ARCHITECTURE, DIGITAL TWIN DASHBOARD & TELEMETRY REPORT

System Type: Industrial IoT Digital Twin	**Frontend Stack:** React 19 + Vite 8 + Tailwind CSS
Cloud IoT: ThingSpeak REST API (#3470506)	**Telemetry Rate:** 10s Continuous Heartbeat Poller

1. Executive Summary & Software Scope

This software documentation presents the complete digital architecture, frontend framework design, real-time telemetry ingestion pipelines, safety state evaluation engine, and analytics visualization platform for the **AI-Powered Smart Waste Segregation Bin -- Digital Twin System**. Built as a production-quality Industrial IoT (IIoT) control center, the software functions as a high-fidelity Digital Twin that renders live fluid capacity levels across multiple physical bin compartments, monitors critical safety parameters (flame, combustible gas/smoke, and ferrous metal proximity), and automatically alerts system administrators when hazardous thresholds are exceeded.

[NOTE] Edge AI vs. Digital Twin Telemetry Isolation Guarantee

The software architecture enforces strict decoupling between physical hardware intelligence and web visualization. Optical image classification (PET/Plastic, Dry/Recyclable, Organic) is executed 100% locally on the ESP32-CAM microcontroller edge hardware. The React Digital Twin Dashboard does NOT run synthetic inferences or artificial predictions; it serves exclusively as an authoritative, high-fidelity monitoring and safety control station consuming live cloud telemetry stream.

2. System Architecture & Data Flow

The software pipeline integrates four operational layers: Edge Hardware Sensors, ESP32 Microcontroller, ThingSpeak IoT Gateway, and the React Digital Twin Dashboard.

```
+-----------------------------------------------------------------------------------+
|                                 SMART WASTE BIN HARDWARE                          |
|  +----------------------------+               +----------------------------------+  |
|  |     ESP32-CAM (Edge AI)    |               | Physical Sensors (HC-SR04, MQ2, |  |
|  | Local Vision Classification|                | Flame Detector, Proximity)      |  |
|  +-------------+-------------+               +----------------+----------------+  |
+----------------|---------------------------------------------+------------------+
                 |                                               |
                 +-----------------------+-----------------------+
                                         |
                                         v
                            +---------------------------+
                            | ESP32 Main IoT Controller|
                            +-------------+-------------+
                                         | HTTP POST (JSON / Form)
                                         v
                            +---------------------------+
                            |   ThingSpeak Cloud IoT    |
                            |   (REST Field Channels)   |
                            +-------------+-------------+
                                         | HTTP GET (10s Polling Hook)
                                         v
                            +---------------------------+
                            | React 19 Digital Twin     |
                            | Dashboard (Vite Frontend)|
                            +---------------------------+
```

3. Software Technology Stack & Dependencies

Package	Version	Purpose / Role	Scope
React	^19.2.8	Core Component Engine & UI State	Frontend Framework
Vite	^8.2.2	Module Bundler & Dev Server	Build Tool
Tailwind CSS	^3.4.17	Industrial Neumorphic UI Styling	Styling Engine
Axios	^1.20.0	Promise-based HTTP REST Client	Data Ingestion
Recharts	^3.10.1	SVG Telemetry Time-Series Charts	Analytics
Lucide React	^1.38.0	Industrial Vector Icon Library	UI Icons
Framer Motion	^13.1.1	Fluid Bin Wave Animations	Animations

4. Cloud IoT Communication & ThingSpeak Field Schema

Field	Sensor Name	Monitored Metric	Valid Range	UI Destination
Field 1	Ultrasonic 1	Bin 01 Fill Level	0% - 100%	BIN 01 — PLASTIC
Field 2	Ultrasonic 2	Bin 02 Fill Level	0% - 100%	BIN 02 — RECYCLABLE
Field 3	Ultrasonic 3	Bin 03 Fill Level	0% - 100%	BIN 03 — ORGANIC
Field 4	Flame Sensor	Fire / Spark Hazard	0 = Safe, 1 = Alert	Emergency Banner
Field 5	Gas (MQ-2)	Air Quality / Smoke	0 - 1000+ PPM	Radial Gas Gauge

Field 6	Proximity	Ferrous Metal Detection	0 = No, 1 = Metal	Metal Indicator Card

5. Software Logic, Algorithms & Mathematical Models

Algorithm 1: Distance Calibration Equation

```
Fill % = min(100, max(0, ((Empty_Dist - Current_Dist) / (Empty_Dist - Full_Dist)) * 100))
```

Algorithm 2: Tri-State Global Safety System Rules

• **EMERGENCY STATE:** Triggered if Flame == 1 OR Smoke >= 800 PPM. Halts bin sorting.

• **WARNING STATE:** Triggered if Any Bin Fill >= 80% OR Smoke 300-799 PPM.

• **NORMAL STATE:** Triggered when Flame == 0 AND Smoke < 300 PPM AND All Bins < 80%.

6. UI Components & Directory Hierarchy

The UI comprises modular glassmorphic cards: Header.jsx, SystemStatus.jsx, KPIGrid.jsx, DigitalTwinBin.jsx, SafetyPanel.jsx, SensorHealth.jsx, CollectionStatus.jsx, AlertsPanel.jsx, and Recharts graph modules.

7. Quick Start & Execution Commands

1. Configure .env with ThingSpeak Channel ID (3470506) and Read API Key.

2. Run npm install to install dependencies.

3. Execute npm run dev for local development server.

4. Execute npm run build to produce production bundle in /dist.